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From scenarios to negotiation games: staging the co-evolution of problem and solution spaces

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ABSTRACT: This paper introduces negotiation games as a method for staging and structuring collaboration in sustainability-oriented engineering design. Building on the Staging Negotiation Spaces (SNS) framework, it shows how scenarios can be re-staged as rule-based artefacts that provoke dialogue and alignment across organisational roles. Drawing on a case in scenographic production, the study demonstrates how negotiation games enable stakeholders to surface divergent concerns, reframe challenges, and co-evolve problem and solution spaces through situated alignment.

KEYWORDS: negotiation games, engineering design, case study, collaborative design, problem-solution co-evolution

1. Introduction

Organisations engaging designers or engineers to address sustainability challenges often articulate these challenges as relatively well-defined technical problems. Sustainability ambitions are frequently framed in terms of missing tools, insufficient data, or suboptimal material choices—implying that the task is one of optimisation rather than negotiation. Yet as designers begin to engage with organisational actors representing different roles, responsibilities, and logics, these framings often prove insufficient. What initially appears as a technical problem unfolds as a contested and fragmented problem space shaped by organisational constraints such as cost structures, production schedules, creative ambitions, and legacy workflows. Under such conditions, designing involves navigating not only technical complexity but also institutional negotiation across actors with diverging understandings of value, feasibility, and responsibility.

These conditions challenge conventional engineering design, long oriented toward linear models, technical optimisation, and limited early stakeholder engagement. Although engineering design research increasingly acknowledges that problems and solutions co-evolve, most established design methods remain predominantly solution-oriented (Obieke et al., 2023). This reflects a persistent bias toward early convergence (Andreasen et al., 2015) and offers limited methodological support for problem reframing and stakeholder alignment in the early stages of collaborative design processes (Pedersen, Clausen, & Jørgensen, 2022). As a result, sustainability concerns are often addressed too late, once key decisions have already stabilised around particular problem definitions.

Material artefacts such as sketches, prototypes, and models have long supported reflection and communication in engineering design, and they can also act as intermediary objects that stage dialogue and translate across professional and organisational boundaries (Vinck & Trompette, 2009). Scenarios are particularly valuable in this respect: they visualise possible futures, reveal assumptions, and create a common reference point for discussion (Anggreeni & van der Voort, 2008). Within participatory design (PD), scenarios are treated as dynamic negotiation devices that evolve through use (Pedersen, 2020), yet in engineering design, they are often introduced too late to inform early problem framing.

This paper explores how scenarios can be re-staged as negotiation games - inscriptive artefacts structured through inscribed negotiation mechanics, including prioritisation, token allocation, and forced trade-offs

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that provoke dialogue and negotiation. Unlike design games primarily oriented toward creativity and ideation, negotiation games focus on structuring multi-actor negotiation across roles and organisational levels. Their purpose is not consensus but situated alignment: recognising interdependencies without erasing differences. Drawing on the Staging Negotiation Spaces (SNS) framework (Pedersen, 2020; Pedersen et al., 2022; Pedersen & Dorland, 2025) illustrated in Figure 1, the paper examines how scenarios-as-negotiation-games can be staged to support the co-evolution of problem and solution spaces in collaborative engineering design settings. Empirically, the paper reports on an illustrative case in which external designers—here bachelor-level engineering design students—collaborated with a Nordic public service organisation to explore sustainability in scenographic production. By staging negotiation games that combine scenarios with inscribed mechanics for prioritising, allocating, and conditioning options, the designers enabled cross-role negotiations that reframed an initially technical project brief into a systemic organisational challenge. More broadly, the paper positions negotiation games as a methodological contribution to engineering design organisation and management, offering designers a means to configure, expand, and reframe both problem and solution spaces through structured negotiation.

2. Literature review: scenarios, negotiation, and alignment in collaborative design

Engineering designers today work within organisational settings where sustainability goals intertwine with competing requirements relating to efficiency, aesthetics, production timelines, and cost. Such multi-dimensional contexts generate contested problem spaces that require negotiation rather than optimisation. Although design theory increasingly acknowledges that problems and solutions co-evolve (Dorst & Cross, 2001), mainstream engineering design methods still presuppose stable problem definitions. Tools such as requirement specifications, weighted objectives, and scorecards effectively support convergence once criteria are agreed, but provide little guidance for the early alignment work between diverse stakeholder perspectives needed to shape those criteria. This results in problems being perceived as stable and focus primarily on evaluating alternative solutions (Andreasen et al., 2015; Obieke et al., 2023). Participatory design traditions provide valuable insight into how such mediation can be enacted. Rather than treating design artefacts as neutral representations, PD views them as instruments for collaboration that materialise concerns and enable shared reflection (Brandt et al., 2008). Artefacts such as mock-ups, prototypes, and scenarios invite participants to articulate values, expectations, and constraints. In participatory design, scenarios are dynamic, evolving through dialogue rather than being fixed proposals (Bødker, 2000). By depicting possible futures, scenarios bridge current practices with envisioned alternatives and create a discursive space where actors can negotiate not only solutions but also what counts as the right problem (Anggreeni & van der Voort, 2008; Bødker, 2000).

Building on this insight, design-oriented games have been developed to structure collaborative interaction through roles, rules, and tangible props, surface tacit knowledge, and support reflection-in-action (Vaajakallio & Mattelmäki, 2014). While such games are effective in fostering ideation and imaginative exploration, they are typically oriented toward generating possibilities rather than negotiating trade-offs across organisational roles and constraints. Negotiation games differ in both orientation and analytical function. Rather than supporting ideation, they inscribe rule-based tensions—such as forced prioritisation, token allocation, or other trade-off–forcing mechanics—that structure multi-actor negotiation across organisational roles. In contrast to serious games (Laamarti et al., 2014) or simulation games, which typically aim to train, simulate, or evaluate predefined scenarios, negotiation games foreground the co-evolution of problem and solution spaces by making competing concerns and interdependencies explicitly negotiable. Their purpose is not consensus but situated alignment: a negotiated recognition of interdependencies that enables collective direction without erasing difference. Negotiation games (Pedersen & Dorland, 2025), emerging from the Staging Negotiation Spaces (SNS) framework (Pedersen, 2020), address this gap by providing a rule-based format for staging negotiation across roles and organisational levels. They operationalise the SNS staging moves—interpreting and (re)framing, inscribing, and inviting and facilitating—by translating empirical concerns into negotiable scenarios and embedding deliberative tensions into game mechanics. Negotiation games can be designed from scratch or can transform established methods, such as scenarios, into playable artefacts. By inscribing scenarios with tensions and prioritisation rules, they create structured settings in which actors must articulate concerns, confront trade-offs, and negotiate directionality. This methodological framing resonates with longstanding calls within engineering design management for approaches that operationalise collaboration across organisational boundaries. Bucciarelli (1994)

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conceptualised design as a social negotiation among distinct "object worlds," while Lloyd and Busby (2003) demonstrated how dialogue and artefacts mediate alignment in distributed teams. Negotiation games build directly on these calls by providing a structured, performative means for designers to stage and facilitate such negotiations in practice. Rather than relying on abstract stakeholder analysis, negotiation games enable situated alignment around evolving concerns. In doing so, they extend the engineering designer's role from problem solver to stager of negotiation (Pedersen 2020), capable of configuring both problem and solution spaces through structured interaction.

In this paper, we build on this literature by demonstrating how scenarios, when staged as negotiation games, can support engineering designers in contexts of institutional complexity. Our case study illustrates how negotiation games can structure dialogue, surface conflicting logics, and support iterative reframing of both problem and solution spaces. In doing so, the paper contributes to design organisation, collaboration, and management by positioning negotiation games as a methodological instrument for enabling situated alignment and embedding sustainability in organisational practice.

3. Methods

This study draws on an illustrative case study (Yin, 2018) of a four-month collaborative design project conducted within a large Nordic public service organisation. The case concerns sustainability-related challenges in scenographic production and involved external designers—here bachelor-level engineering design students enrolled in a sustainable design engineering programme—working in collaboration with the organisation's scenography unit.

The empirical material comprises: (1) semi-structured interviews with scenographers, producers, and managers; (2) desk research into organisational workflows, procurement routines, and sustainability-related documentation; and (3) two facilitated workshops in which scenarios and other artefacts were staged as negotiation games. Rather than approaching the case as an evaluation of solutions or organisational outcomes, the analysis attends to how negotiation is staged and enacted in practice. Specifically, it focuses on how scenarios were configured as negotiation games through the three SNS staging moves—*interpreting and (re)framing, inscribing, and inviting and facilitating* (see Figure 1)—and how role positions shaped negotiations across problem and solution spaces.

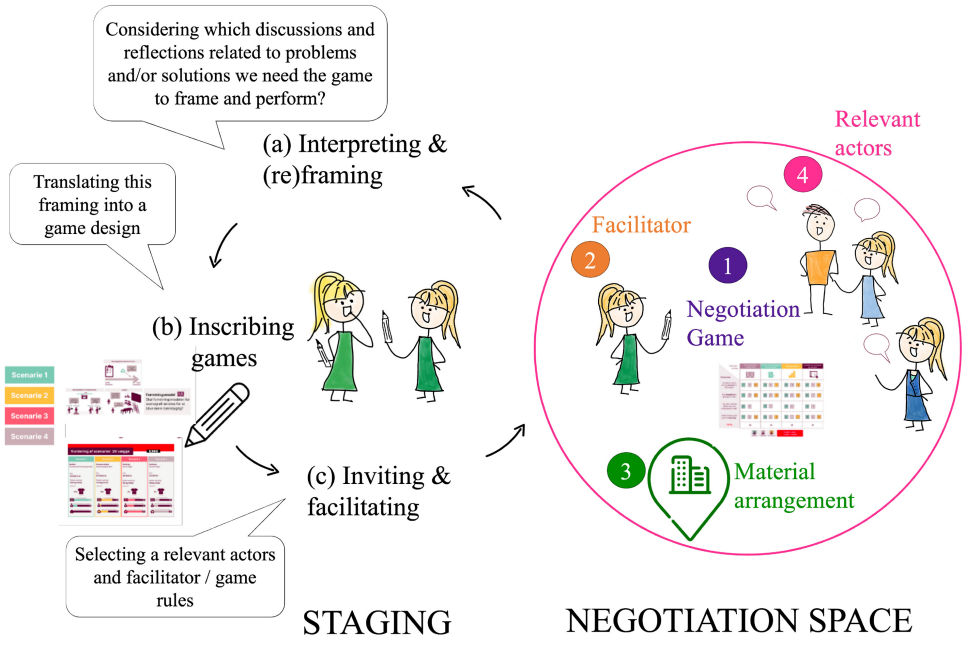


Figure 1. Staging moves (figure adapted from Pedersen & Dorland, 2025)

4. Findings: staging and negotiating problems and solutions

The findings are structured by the three SNS moves—interpreting, inscribing, and inviting—to show how scenarios were transformed into negotiation games and how role positions shaped negotiation outcomes.

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4.1. Workshop 1 - negotiating barriers to sustainable scenography

4.1.1. Interpreting

The organisation's sustainability coordinator initially framed the design brief around material knowledge and the development of a CO_2-calculation tool, suggesting a predominantly technical understanding of the task. To investigate and potentially broaden this framing, the designers conducted a series of interviews across the scenography unit. A producer, a line producer, a production designer and a production assistant described their daily practices, frustrations, and ambitions. These conversations were supplemented with desk research into tendering processes, procurement routines, and the organisation's sustainability strategies.

As part of the interpreting move, the designers constructed an affinity diagram in which statements and observations were clustered into thematic categories. From this analysis, six recurring barriers to sustainable scenography were identified:

1. **Limited knowledge of sustainable materials** – lack of guidance on environmentally preferable materials.
2. **Sustainability framed as a constraint of creativity** – concerns that rules or standards would limit creative freedom and weaken the organisation's distinctive visual identity.
3. **Late integration of sustainability** – sustainability was often discussed only after concepts and budgets were finalised, making changes costly or impractical.
4. **Logistical bottlenecks** – limited storage capacity and the carbon cost of transport constraining material reuse.
5. **Short-term production-use cycles** – scenographic sets designed for single productions and subsequently discarded despite remaining usability.
6. **Competitive pressures** – internal market dynamics, where productions could turn to external suppliers, encouraged fast and cheap delivery at the expense of sustainability.

Although the initial brief centred on material knowledge (barrier #1), the designers' interpretation suggested a more complex and interconnected problem space. Sustainability appeared less as a technical deficit and more as a systemic organisational challenge spanning creative, operational, and logistical domains. This reframing motivated the designers to inscribe the six barriers into scenarios capable of staging negotiation across roles rather than merely communicating analytical findings.

4.1.2. Inscribing: from barriers to a prioritisation-based negotiation game

Translating the six barriers into a rule-based prioritisation game constituted the SNS inscription move. Barriers and concerns were materialised as scenarios in the form of playable artefacts, whose rules rendered trade-offs explicit. The rules required participants to collectively rank the scenarios from most to least critical, and equal rankings were prohibited to ensure that trade-offs became explicit.

Each card provided a short, accessible description of a barrier, situated in a concrete setting with actors, activities, and temporal context, and accompanied by an illustration. For example, one scenario featured a carpenter and a technical coordinator discussing which wall panels to use for an upcoming production set. The carpenter suggests plywood because it is quick and familiar. The coordinator remembers that sustainability is now a priority but has no clear guidelines to rely on. With the deadline only days away, they decide on the cheapest and fastest option—accompanied by an illustration of the two actors in a workshop with speech bubbles.

4.1.3. Preparing the ground: a warm-up as pre-negotiation

Before engaging with the barrier scenarios, the workshop began with a short warm-up game. Participants were asked to prioritise three overarching concerns identified in the interviews—time, economy, and sustainability. Although simple in format, the warm-up was deliberately staged to foreground tensions and prepare participants to articulate and negotiate diverging perspectives before confronting the more complex scenarios.

4.1.4. Inviting and facilitating

The workshop was staged to include participants from different organisational roles and levels, while deliberately excluding the sustainability coordinator who had authored the original brief. A carpenter was invited to contribute hands-on knowledge of materials and construction processes. A team leader represented collective scenographic concerns and the balance between creative ambition and operational feasibility. A manager was included to ensure that strategic considerations and decision-making authority

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were present. This constellation was designed to anchor ownership across the organisation while staging a cross-level negotiation of sustainability barriers.

4.1.5. Negotiation dynamics

During the warm-up game, a manager introduced a fourth concern—uniqueness—describing it as the organisation's defining creative asset. This addition effectively inscribed creative ambition into the negotiation space, reshaping the logic of the warm-up and explicitly placing sustainability in tension with aesthetic distinctiveness. The intervention illustrated how negotiation games are performative rather than static: they absorb emergent concerns and reconfigure the conditions for dialogue.

As participants moved to the main ranking exercise, role-based differences became increasingly visible. Team leaders emphasised creative freedom and the importance of delivering a visual identity often already embedded in the script, aligning with concerns expressed in scenario #2. Carpenters highlighted the constraints imposed by tight deadlines, noting how time pressure limited opportunities for experimenting with material reuse, particularly when combined with competitive pressures (scenarios #3 and #6). Managers and carpenters alike stressed the close coupling of time and economy, pointing out that efficient task execution translated directly into revenue.

Toward the end of the session, a senior manager reframed the discussion: "*These issues are symptoms of a bigger problem—the business model we work under.*" This remark prompted the collective articulation of a seventh scenario: the business model constraint. Although not originally inscribed by the designers, the scenario emerged through the negotiation itself, demonstrating how the game mechanics enabled participants to generate new framings of the problem space.

4.1.6. Outcome

The final ranking prioritised late integration of sustainability (scenario #3) as the most critical barrier, followed by creative constraints, logistical bottlenecks, and the newly introduced business model scenario. Material knowledge (scenario #1), which had structured the original brief, was ranked significantly lower. Through the negotiation, sustainability was reframed from a question of technical expertise to a systemic organisational challenge. Managers and carpenters consistently emphasised the coupling of time and economy, while team leaders foregrounded creative freedom and uniqueness as non-negotiable conditions. These positions clarified why sustainability concerns were typically introduced late in the process: early stages were dominated by pressures for efficiency and distinctive expression, leaving little room for alternative practices. The spontaneous inclusion of the business model constraint further shifted the framing away from technical fixes toward structural organisational issues. Rather than resolving disagreement, the negotiation produced a form of situated alignment around what was at stake. Sustainability could no longer be advanced through improved material knowledge alone but required negotiations across creative, operational, and managerial domains.

4.2. Workshop 2 – negotiating solution pathways

4.2.1. Interpreting and (re)framing

Building on the outcomes of Workshop 1, the designers interpreted the ranking and accompanying discussions as evidence that sustainability challenges in scenographic production were systemic rather than technical. The repeated emphasis on timing suggested that sustainability needed to be integrated earlier in the production process, while the spontaneous articulation of the business model constraint underscored the structural nature of the problem. Rather than treating these findings as conclusions, the designers used them as a basis for staging a second negotiation space focused on possible solution pathways. From this interpretation, four solution-oriented scenarios were developed, each addressing barriers that had been ranked as particularly critical in Workshop 1: (1) reconfiguring the business model, (2) introducing modular scenographic systems, (3) increasing sustainability requirements in workflows, and (4) developing a digital inventory system. The scenarios were designed not as proposals to be selected, but as negotiable directions whose implications and interdependencies could be explored.

4.2.2. Inscribing: a token-based negotiation game

The four solution scenarios were inscribed into a token-based negotiation game, constituting the SNS inscription move. As in Workshop 1, each scenario was presented on a card with a concise description, concrete examples, and potential implications. The mechanics of the game, however, differed in order to

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shift the negotiation from problem identification to evaluation of pathways. Participants were given a set of fixed-value tokens (1–4), which they were required to allocate across four criteria: relevance, complexity, time savings, and uniqueness. Relevance captured whether a solution applied broadly across roles or primarily to specific functions. Complexity was used to identify perceived low-hanging fruits. Time savings and uniqueness were carried over from Workshop 1 to maintain continuity in the negotiation logic. Crucially, all tokens had to be assigned, forcing participants to make explicit trade-offs rather than distributing attention evenly across scenarios.

By combining scenario cards with mandatory token allocation, the game mechanics made both preferences and priorities visible, while simultaneously exposing tensions between immediate feasibility and longer-term structural change.

4.2.3. Inviting and facilitating

The workshop was staged with the same cross-section of organisational roles as Workshop 1. A carpenter contributed hands-on production knowledge, a team leader represented scenographic and creative concerns, and a manager ensured that strategic and organisational perspectives were present. This constellation was intended to test whether the previously negotiated problem framings could be translated into viable solution pathways, while anchoring ownership across operational, creative, and managerial domains.

4.2.4. Negotiation dynamics

As participants engaged with the four scenarios, their token allocations reflected role-based perspectives. The carpenter consistently rated the business model scenario lowest, explaining that it lay outside his immediate field of responsibility and was therefore difficult to relate to in practical terms. In contrast, modular scenography was scored highly, as it represented a tangible change that could be implemented within existing production practices. The team leader similarly valued modularity but stressed that it would only be viable if it preserved creative freedom and allowed for distinctive visual expression. The manager prioritised the business model scenario, arguing that without structural adjustments to key activities and resource allocations, none of the other solutions could be realised.

While the initial allocations highlighted divergent priorities, the negotiation gradually shifted toward interdependencies between the scenarios. Participants collectively recognised that modular scenography would require additional storage capacity, revised workflows, and potentially new roles—none of which were feasible without changes to the business model. Similarly, raising sustainability requirements in workflows was seen as unrealistic unless systemic incentives supported earlier integration of sustainability considerations. The digital inventory scenario, though initially valued by the team leader, was deprioritised as discussions progressed. Participants noted that ongoing internal initiatives were already addressing aspects of digital overview, and the scenario was increasingly framed as a supportive rather than transformative intervention, with limited strategic impact compared to the others.

4.2.5. Outcome

The negotiation revealed a tension between immediate relevance and long-term feasibility. Although modular scenography and digital overview initially scored highly due to their apparent practicality, participants ultimately aligned around the view that reconfiguring the business model constituted a structural precondition for making any of the other solution pathways viable. As one participant noted, "*we cannot introduce modular systems unless the business model changes to make room for storage and new tasks*." The outcome of the workshop therefore prioritised business model reconfiguration as the central direction, with modular scenography and adjusted workflows positioned as dependent and supporting elements. The digital inventory system was set aside as a secondary concern. In this way, Workshop 2 consolidated and extended the reframing initiated in Workshop 1. Sustainability was no longer negotiated as a matter of isolated solutions, but as a question of organisational conditions that enabled or constrained action. The negotiation game format allowed participants to acknowledge role-based differences while still achieving situated alignment around a shared direction—without requiring consensus or compromise.

4.3. Translating insights into circulating organisational artefacts

Following the second workshop, the designers stepped back from the negotiation setting to consolidate the outcomes and stabilise the emerging alignment in forms that could travel beyond the staged negotiation space. Rather than treating the prioritised solution pathways as strategies to be implemented,

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they translated the reframed problem understandings and negotiated solution pathways into material artefacts capable of circulating within the organisation. Specifically, the designers inscribed the prioritised solution scenarios into Business Model Canvases (BMCs) and conducted comparative Life Cycle Assessments (LCAs). The LCAs modelled wood flows across 20 productions under different reuse strategies, comparing disposal, recycling, and internal or external reuse. The analysis showed that reuse strategies outperformed recycling after as few as five cycles, even when additional transport was taken into account. Importantly, these artefacts were not introduced as neutral evaluation tools. Instead, they functioned as inscriptive devices that extended the negotiations by embedding both the expanded problem space and the conditioned solution space in forms that could move between actors, projects, and organisational settings. In doing so, the BMCs and LCAs carried the negotiated alignment forward by articulating sustainability as a systemic design concern linking material choices, workflows, and business models. Rather than closing the negotiation, the artefacts enabled its continuation in other organisational contexts.

5. Discussion: staging moves and role-based alignment

This study has examined how negotiation games can be used as staging devices for organising collaboration in engineering design contexts characterised by institutional complexity and competing concerns. While participatory design traditions have long employed design games to support ideation, creativity, and dialogue (Brandt et al., 2008; Vaajakallio & Mattelmäki, 2014), the findings presented here demonstrate a different orientation. Negotiation games are not primarily concerned with creativity or consensus-building, but with structuring negotiation around concerns, trade-offs, and interdependencies across organisational roles. In this sense, they extend familiar design artefacts—such as scenarios, prioritisation matrices, and requirements specifications—not by replacing them, but by transforming their role, sequence, and function within the design process.

Rather than assuming a linear progression from problem definition to solution generation, the case shows how negotiation games enable the co-evolution of problem and solution spaces through staged negotiation. Figure 2 synthesises this contribution by illustrating how negotiation games mediate between problem spaces and solution spaces through situated alignment, rather than resolving complexity through early convergence.

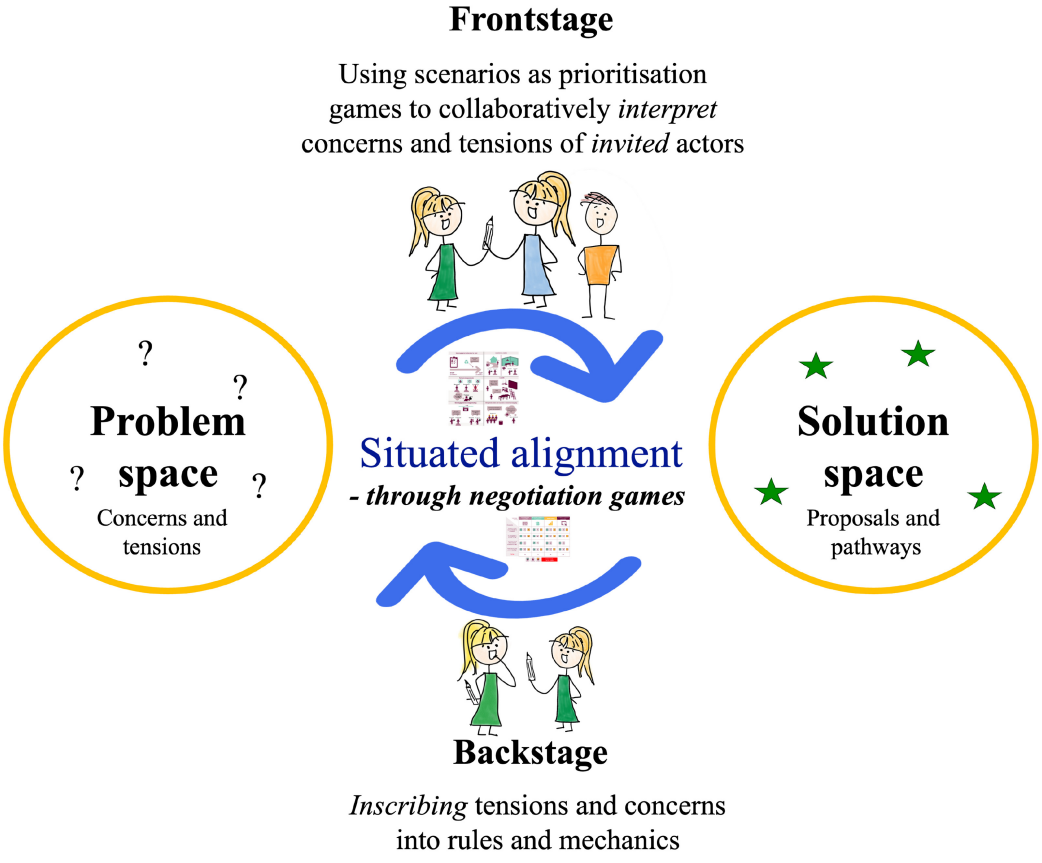


Figure 2. Negotiation games mediate the co-evolution of problem and solution spaces by staging situated alignment across organisational roles

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5.1. Negotiating problem framing

In engineering design, problem definition is often treated as a preliminary step that stabilises early and remains largely unchanged as development progresses. The findings challenge this assumption by showing how negotiation games stage problem framing itself as a matter for negotiation. In Workshop 1, empirical material from interviews and desk research was translated into barrier scenarios that rendered sustainability concerns negotiable across organisational roles. This interpretive work was strategic rather than neutral: by moving beyond the sustainability coordinator's initial focus on material knowledge, the designers reframed sustainability as a systemic organisational concern involving creative, operational, and logistical dimensions.

Crucially, this reframing did not occur through analytical argument alone. By requiring participants to prioritise and compare scenarios, the negotiation game enabled problem framing to unfold as a collective and performative accomplishment. The spontaneous introduction of the business model scenario illustrates how negotiation games allow problem spaces to expand dynamically in response to role-based concerns, rather than being fixed in advance.

5.2. Embedding tensions through game mechanics

A key contribution of negotiation games lies in their inscriptional logic. Across the two workshops, tensions were embedded directly into the game mechanics through forced rankings and fixed-value tokens. These rules compelled participants to make trade-offs explicit, rendering differences in priorities unavoidable. Unlike open discussion formats, where disagreement can remain implicit or be smoothed over, negotiation games materialise tensions as part of the process.

The mechanics also created room for emergence. New concerns—such as uniqueness and the business model constraint—entered the negotiation not despite the structure of the game, but because of it. In this way, negotiation games operate not merely as intermediary objects (Vinck, 2012), but as deliberative devices that actively structure how concerns are articulated, compared, and reconfigured.

5.3. Role-based negotiation and situated alignment

Who is invited into the negotiation, and under what conditions, proved central to the alignment work observed. By assembling participants from operational, creative, and managerial roles—and by structuring their interaction through shared artefacts and rules—the negotiation games redistributed influence away from formal hierarchy and toward articulated concerns. The result was not agreement but situated alignment: a negotiated recognition of interdependencies that enabled collective direction without erasing difference.

Traditional design management tools often assume that convergence and consensus are necessary outcomes. The findings instead show that negotiation games decouple alignment from consensus. Divergent priorities—creative uniqueness, operational feasibility, and systemic reform—were not resolved but made mutually intelligible and conditionally related. Alignment emerged through recognising what had to change first for other options to become viable.

5.4. From negotiation spaces to organisational circulation

The final contribution of the case concerns what happens after negotiation spaces close. The findings show how reframed problem understandings and conditioned solution pathways can be inscribed into material artefacts—such as Business Model Canvases and Life Cycle Assessments—that are capable of travelling and circulating within organisations. These artefacts do not implement decisions or resolve negotiations. Rather, they stabilise and extend situated alignment by embedding negotiated insights in forms that can move between actors, projects, and organisational settings.

Through this inscription, sustainability was articulated as a systemic design concern linking material choices, workflows, and business models, rather than as an isolated technical issue. In this way, negotiation games not only mediate between problem and solution spaces within staged settings but also support the translation of negotiated insights back into the organisational arena.

5.5. Implications for engineering design organisation and management

In practical terms, negotiation games extend the repertoire of engineering design management by complementing established evaluation-oriented methods. Tools such as requirements lists, weighted

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objectives, and scorecards have long supported collaboration by quantifying criteria, ranking alternatives, and converging on a single "best" solution. These tools are effective once criteria are stabilised, but they typically assume that problem framings are fixed and that trade-offs can be predefined. As a result, they risk narrowing complexity too early and leaving limited room for contested concerns to be negotiated.

Negotiation games share with such methods a concern for explicit prioritisation but differ in orientation and timing. Rather than treating prioritisation as a final evaluative step, negotiation games bring it into the framing of both problems and solutions. By embedding tensions into scenarios and rules, they hold problem and solution spaces open long enough for competing perspectives to be articulated, compared, and reconfigured. Instead of aiming for consensus around a single optimal option, they support situated alignment by making interdependencies visible without erasing difference.

Importantly, negotiation games should not be understood as replacements for established design management tools. Rather, they create the conditions under which such tools can later be meaningfully applied. Once negotiated alignment has been staged, more conventional instruments may be reintroduced as mechanisms of convergence grounded in prior negotiation rather than assumed agreement.

6. Conclusion

This paper has examined how scenarios, when re-staged as negotiation games through the Staging Negotiation Spaces (SNS) framework, can support sustainability-oriented engineering design in contexts characterised by institutional complexity and competing organisational concerns. Based on an illustrative case from scenographic production in a Nordic public service organisation, the study traced how rule-based negotiation games enabled designers to reframe an initially technical project brief and to negotiate conditioned directions for organisational change across operational, creative, and managerial roles.

The paper contributes to design organisation and management through three interrelated methodological advances. First, it shows how scenarios can be transformed into negotiation games by inscribing them with rule-based mechanics—such as prioritisation, token allocation, and forced trade-offs—that make concerns and interdependencies explicitly negotiable. Rather than functioning as representations of possible futures, scenarios become deliberative devices that structure how organisational actors articulate, compare, and condition what is at stake.

Second, the paper demonstrates how negotiation games support the co-evolution of problem and solution spaces. Across the two workshops, problem framings and solution pathways were not stabilised in advance but emerged through successive rounds of negotiation, where reframings in one space directly conditioned what could be explored in the other. In this way, negotiation games enabled designers to hold complexity open long enough for organisational constraints, creative ambitions, and sustainability considerations to be explored as mutually shaping rather than sequentially resolved.

Third, the study advances the concept of situated alignment as an alternative to consensus. The findings show how negotiation games redistribute influence away from formal hierarchy and toward articulated concerns by embedding negotiation in shared artefacts and rules. Alignment emerged not through agreement, but through a negotiated recognition of interdependencies—for example, how creative uniqueness, operational feasibility, and business model conditions were rendered mutually conditional without being reconciled into a single perspective.

Taken together, these insights position negotiation games as a methodological contribution that complements established engineering design tools. Rather than replacing requirements specifications, evaluation matrices, or scorecards, negotiation games create the conditions under which such instruments can later be meaningfully applied. By staging negotiation through inscribed mechanics that make trade-offs unavoidable, negotiation games enable engineering designers to engage productively with organisational complexity, support the co-evolution of problem and solution spaces, and embed sustainability considerations within ongoing organisational practice.

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